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Final Terms Spring 2020

Current M.C.Q

1. One nanometer is equal to _____ meter. **1 billionth**

2. Generally nanotechnology deals with structures sized between _____nanometer in at least one dimension. **1 to 100**

3. Who developed Humulin? **Genentech**

4. In clinical trials, lymphotropic iron oxide nanoparticles acted as effective contrast agents and allowed the detection of small nodal metastases in men with_____ **prostate cancer**

5. Nanorobots are nanodevices have a diameter of about _____ **0.5 to 3 microns**

6. Fullerenes and bucky balls has_____ **anti-oxidant and smoothing properties**

7. The US Food and Drug Administration has already approved first generation nanodrugs such as Abraxane, a nanoformulation of_____ **anticancer chemotherapy paclitaxel**

8. AuNPs _____ show relatively high-scattering properties making them more applicable for biomedical applications. **20–80 nm**

9. The random motion of molecule is due to_____ **Kinetic energy**

10. The energy separation between the valence band and the conduction band is called____ **Eg**

11. Biomimetic materials are materials developed using inspiration from_____ **Nature**

12. _____ is the study of the formation, structure, or function of biologically produced substances and materials and biological mechanisms. **Biomimetics and bionics**

13. Diatoms are a large group of eukaryotic microalgae that have a rigid cell wall made of_____ **Silica (SiO_2)**

14. A_____ is a fourth state of matter. **Liquid crystal**

15. Conductive polymers have revealed that these are made of a sequence of metallic nanoparticles about _____in diameter. **10 nm**

16. Nanofibers are fiber with diameter in nanometer range_____ **50 to 1000**

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17. Ceramic materials by definition are ionically bound are included in this category are_____ **Al_2O_3, Si_3N_4, MgO, SiO_2, Na_2O, CaO and ZrO_2**

18. A new allotrope of carbon was discovered formed of 60 atoms of carbons linked together through single covalent bonds arranged in a highly symmetrical are named as _____ **buckyball**

19. Some nanotubes have a property called chirality, an expression of___ **longitudinal twisting**

20. The temperature stability of carbon nanotubes is estimated to be up to_____ **2800°C**

21. _____ are materials where a polymer is filled with an inorganic synthetic and/or natural compound in order to increase several properties, such as heat resistance or mechanical strength, or to decrease other properties, such as electrical conductivity. **Polymer composites**

22. _____ are those coatings that are applied to the surface of a component in order to control its friction and wear. **Tribological coatings**

23. A 300 nm ITO coating on glass provides more than ____transmission for the wavelengths predominant in sunlight. **80 %**

24. The resolving power of an optical microscope is around_____ **0.2 µm**

25. A_____ is a device that detects and responds to some type of input from the physical environment. **Sensor**

26. A_____ sensor converts light rays into an electronic signal. **Optical**

27. The optical properties of noble metal nanoparticles are dominated by an effect_____ **localized surface Plasmon resonance (SPR).**

28. Which is the part of a biosensor that processes the transduced signal and prepares it for display? **Electronics**

29. Which one of the following is physicochemical component? **Enzyme**

30. Biosensors can be utilized to monitor the presence of_____ **products, biomass, enzyme, antibody or by-products of the process**

31. _____ is the nitty-gritty of crucial functions such as (signalling, transport and precursor in biosynthesis of nucleic acids, amino sugars and proteins). **Glutamine**

32. About _____of the population worldwide suffers from diabetes. **3%**

33. Sensors are now being used for the identification of toxic chemical compounds at____ **ppm and ppb**

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34. Which one of the following is Antinutritional compound? **oxalate and glycoalkaloids**

35. The microarray technique is based on _____of DNA. **Hybridization**

36. In the body, cells are supported in their growth and function by a natural scaffold, called_____ **extracellular matrix**

37. Which one of the following is not a diagnostic imaging technique? **Multiscan**

38. Iron oxide and cobalt oxide with a mean size of _____can be encapsulated in these nanocavities. **6 nm**

39. Nano gold has_____ **healing and anti-oxident properties** **(repeat 2 times in my paper)**

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Subjective

2 marks

Qno1: Define Bioreceptor.

A molecule that specifically recognizes the analyte is known as a bioreceptor. Enzymes, cells, aptamers, deoxyribonucleic acid (DNA) and antibodies are some examples of bioreceptors.

Qno2: What are two types of glucose detection biosensor?

First-generation of glucose biosensors:

These sensors were initially based on an electrochemical approach, which used the enzyme glucose oxidase (GOx).Electrochemical sensors were chosen for blood-glucose measurements due to their high sensitivity, on the order of μM to mM, good reproducibility and ease of fabrication at relatively low cost.

Second-generation of Glucose Biosensors:

Limitations of the first generation glucose biosensors were overcome by using mediated glucose biosensors, second generation glucose sensors. The improvements were achieved by replacing oxygen with non-physiological electron acceptors, called redox mediators that were able to carry electrons from the enzyme to the surface of the working electrode.

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Qno3: Define Bioprospecting.

Bioprospecting can define as rich and valuable sources of new genes, proteins and metabolic processes with important applications for human benefits.

Qno4: Explain the Linearity of Biosensor with mathematical equation.

Linearity is the attribute that shows the accuracy of the measured response (for a set of measurements with different concentrations of analyte) to a straight line. Linearity of the biosensor can be associated with the resolution of the biosensor and range of analyte concentrations under test.

Mathematical Equation:

y = mc

y is the output signal

m is the sensitivity of the biosensor

c is the concentration of the analyte

Qno5: Describe the applications of Biosensor in food industry.

Applications of Biosensor in food industry:

1. Quality and safety, maintenance of food products and processing.
2. Food authentication and monitoring with objective and consistent measurement of food products, in a cost effective manner, are desirable for the food industry.
3. Biosensors are used for the detection of pathogens in food.
4. Monitoring of ageing of beer using enzymatic biosensors, based on cobalt phthalocyanine – during storage.

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3 marks

Qno1: Describe the applications of nanoparticles in remediation and mitigation.

Applications of nanoparticles in remediation and mitigation:

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1. The use of zero-valent (Fe0) iron nanoparticles for them remediation of contaminated groundwater and soil is a good example of how environmental remediation can be improved with nanotechnology.

2. When exposed to air, iron oxidises easily to rust; however, when it oxidises around contaminants such as trichloroethylene (TCE), carbon tetrachloride, dioxins, or PCBs, these organic molecules are broken down into simple, far less toxic carbon compounds.

3. Some industries have started using an 'iron powder' to clean up their new industrial wastes.

Qno2: Write 3 applications of Self-assembled monolayers (SAMs).

Applications of Self-assembled monolayers (SAMs):

1. SAMs are extremely useful in thin-film technology since they provide a simple, versatile and relatively inexpensive way of producing coatings with specific functionalities.

2. Thin coatings are used in many applications, for example to control surface wetting and adhesion properties; to provide chemical resistance or specific biocompatibility; to confer antibacterial properties; and for sensors.

3. SAMs are particularly interesting for developing DNA and protein micro and nanoarrays for medical screening and for future electronic components in computer chips and other ICT developments.

Qno3: What is spectroscopy? Describe its principle.

Spectroscopy is defined as the branch of science that is concerned with the investigation and measurement of spectra produced when matter interacts with or emits electromagnetic (EM) radiation.

Principle:

Spectroscopy is a technique that uses the interaction of energy with a sample to perform an analysis. Depending on the wavelength of the electromagnetic used and the type of interaction with matter that occurs (absorption, scattering, etc.), different spectra are measured from which much information can be inferred.

Qno4: Define Nanotechnology and highlight its areas of applications.

Nanotechnology literally means any technology on a nanoscale that has applications in the real world. Nanotechnology is group of emerging technology in which the structure of matter is

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controlled at the nanometer scale to produce novel materials and device that have useful and unique properties.

Areas of applications:

1. Information Technology

2. Mechanical Eng. & Robotics

3. Biotechnology

4. Transportation

5. National Security & Defense

Qno5: What is difference between super hydrophilic and super hydrophobic coating?

Superhydrophilic coating	Superhydrophobic coating
Photocatalytic coatings (commercialized as self cleaning glass) which use the catalytic properties of titanium dioxide (TiO_2).	Nanocoating is hydrophobic (water repellent), oleo phobic (oil repellent) surface layer that repels water, oil, dirt, and other dry, particles.
When irradiated with UV light, the coating becomes superhydrophilic: therefore, rain water adheres to the glass providing 'selfcleaning'.	High hydrophobicity eco-friendly nano coating can be applied onto objects to make them waterproof for the long-term.
Superhydrophilic coatings are also useful for rendering a surface fog-resistant.	The opposite of superhydrophilc coatings are superhydrophobic coatings, which totally repel water.

Qno6: What are the applications of carbon nanotubes?

Particularly, in the area of nanoelectronics, CNTs and graphene nanoribbons (GNRs) demonstrates wide range of applications such as

1. Energy storage devices

2. Energy conversion devices that includes thermoelectric

3. Photovoltaic devices

4. Field emission displays and radiation sources

5. Nanometer semiconductor transistors

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6. Nanoelectromechanical systems (NEMS

7. Electrostatic discharge (ESD) protection

8. Interconnects and passive

5 marks

Qno1: Describe Bioreceptor in detail.

A molecule that specifically recognizes the analyte is known as a bioreceptor. Enzymes, cells, aptamers, deoxyribonucleic acid (DNA) and antibodies are some examples of bioreceptors. The process of signal generation (in the form of light, heat, pH, charge or mass change, etc.) upon interaction of the bioreceptor with the analyte is termed bio-recognition.

Qno2: What are different intrinsic and extrinsic factors that influence of Self-assembly of nano materials?

Self-assembly is commonly controlled by certain intrinsic material parameters e.g., composition, strain, thickness, phase transformation, structural changes and results from the interaction between different factors e.g., deposit/substrate, liquid/gas/solid phases, crystals.

A number of extrinsic factors, including thermal treatment, chemical and electrochemical reactions, mechanical stress, electric or magnetic fields can strongly influence the self-assembled morphologies.

Qno3: What are the potential benefits of nanocoating?

Some benefits of nanocoatings are:

1. Reduced need for substrate maintenance activities to prevent corrosion

2. Reduced labor and material costs

3. Prolonged substrate lifespan and application use

4. Increased substrate resistance to temperature fluctuations

5. Substrate waterproofing

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6. Ultraviolet (UV) stability and abrasion resistance

Qno4: Write down five ways of nanotechnology impacts on our lives daily.

1. Faster, smaller, and more powerful computers that consume farm less power, with longer-lasting batteries

2. Faster, more functional, and more accurate medical diagnostic equipment. Lab-on-a-chip technology enables point-of-care testing in real time, which speeds up delivery of medical care. Nanomaterial surfaces on implants improve wear and resist infection.

3. Nanoparticles in pharmaceutical products improve their absorption within the body and make them easier to deliver, often through combination medical devices. Nanoparticles can also be used to deliver chemotherapy drugs to specific cells, such as cancer cells.

4. Nanoparticles or nanofibers in fabrics can enhance stain resistance, water resistance, and flame resistance, without a significant increase in weight, thickness, or stiffness of the fabric. For example, “nanowhiskers” on pants make them resistant to water and stains.

5. Water filters that are only 15-20 nanometers wide can remove nanosized particles, including virtually all viruses and bacteria. These costefficient, portable water treatment systems are ideal for improving the quality of drinking water in emerging countries.

Qno5: Explain in detail classification of sensor on the base of transducer.

Optical sensors

An optical sensor converts light rays into an electronic signal. The purpose of an optical sensor is to measure a physical quantity of light and, depending on the type of sensor, and then translates it into a form that is readable by an integrated measuring device. Optical sensors can be either internal or external. External sensors gather and transmit a required quantity of light, while internal sensors are most often used to measure the bends and other small changes in direction.

Electrochemical Biosensor

Electrochemical Biosensors is a simple device.

1. It measures the measurement of electronic current, ionic or by conductance changes carried by bio-electrodes.

2. These sensors employ redox reactions to quantify the amount of an analyte.

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3. The current flowing through the system or the potential difference between the electrodes as a result of the oxidation and reduction reactions involving the analyte are used for its quantification in the sample.

Cantilever biosensor

A cantilever biosensor is a biosensor made of numerous arms, which are tens of micrometres long but very thin.

1. The surface of the cantilever is functionalized with a nanometre-thick layer of coating which ensures anchorage of the probe material (which can be a DNA strand or a protein, for example).

2. Each cantilever is different and can probe for a different target.

3. In this type of sensor, the adsorption of the analyte to the specific targets on a cantilever causes a surface stress and bends the cantilever.

Plasmonic biosensors

1. It is used to detect the binding of biological molecules onto arrays of probe biomolecules covalently attached to chemically-modified gold surfaces.

2. The optical properties of noble metal nanoparticles have received significant research attention in recent years for their potential as components in many applications, including chemical/biochemical sensors.

3. The optical properties of noble metal nanoparticles are dominated by an effect called localized surface Plasmon resonance (SPR).

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10 marks

Qno1: Define the following terms:

a. **Ampermetric Biosensor**

The Biosensors are based on the electrons movement, i.e. electronic current determination as a reaction of enzymecatalyzed redox reaction.

b. **Blood Glucose Biosensor**

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The blood glucose Biosensors are used widely throughout the world for diabetic patients. It has single use disposable electrodes with glucose oxide and derivatives of a mediator (Ferrocence) and the shape of the blood glucose Biosensor looks like a watch pen.

c. **Potentiometric Biosensor**

In this type of Biosensors changes the concentration of ionic is determined by the ion-selective electrodes in this pH electrodes are used most commonly.

d. **DNA biosensors**

In the future, DNA will find use as a versatile material from which scientists can craft biosensors. DNA biosensors can theoretically be used for medical diagnostics, forensic science, agriculture, or even environmental clean-up efforts.

e. **Calorimetric Sensors**

Calorimetric sensors (Note the spelling; Colorimetric sensors will refer to optical sensors!!), involve the measurement of heat that is generated (Calorie: unit of heat; Calorimetric: Measurement of heat). These sensors typically utilize thermistors that transform heat generated or loss during a reaction into an electrical signal.

Qno2: Describe the applications of Biosensor.

Biosensors are devices comprising a biological element and a physiochemical detector that are used to detect analytes.

1. These instruments have a wide range of applications ranging from clinical through to environmental and agricultural.

2. The devices are also used in the food industry.

3. Some examples of the fields that use biosensor technology include:

4. General healthcare monitoring

5. Screening for disease

6. Clinical analysis and diagnosis of disease

7. Veterinary and agricultural applications

8. Industrial processing and monitoring

9. Environmental pollution control

In food processing, monitoring, food authenticity, quality and safety

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1. Quality and safety, maintenance of food products and processing

2. Food authentication and monitoring with objective and consistent measurement of food products, in a cost effective manner, are desirable for the food industry.

3. Thus development of biosensors in response to the demand for simple, real-time, selective and inexpensive techniques is seemingly propitious.

4. Monitoring of ageing of beer using enzymatic biosensors, based on cobalt phthalocyanine – during storage.

In fermentation processes

1. In fermentation industries, process safety and product quality are crucial.

2. Thus effective monitoring of the fermentation process is imperative to develop, optimize and maintain biological reactors at maximum efficacy.

3. Biosensors can be utilized to monitor the presence of products, biomass, enzyme, antibody or by-products of the process to indirectly measure the process conditions.

4. Biosensors precisely control the fermentation industry and produce reproducible results due to their simple instrumentation, formidable selectivity, low prices and easy automation.

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